Análisis Exploratorio

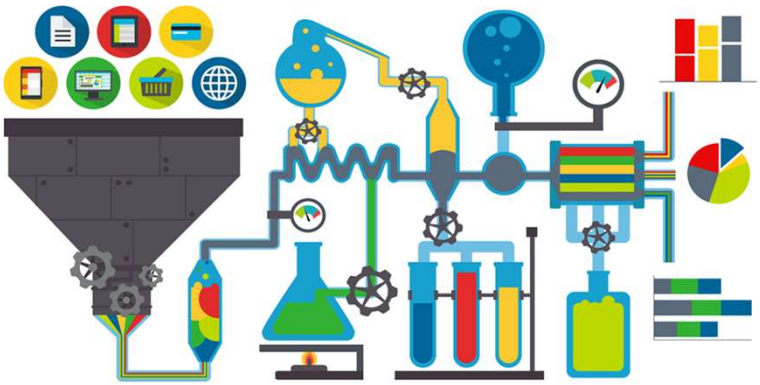
RETO DE LA ORGANIZACION

PREGUNTA SMART
¿Cómo?
¿Cuáles?
¿Cuándo?
¿Por qué? *

DATOS:
Materia prima.

MODELO/ DASHBOARD
- Implementación y ajuste del modelo
- Dashboard?

EVALUACION:
Resuelve la pregunta?
Validación criterios de éxito.



DESPLIEGUE:
- Portal, empresa
- Modificación proceso
- Recurso humano

ANALISIS EXPLORATORIO:
Limpieza, preparación, transformación.
Selección de variables (atributos).

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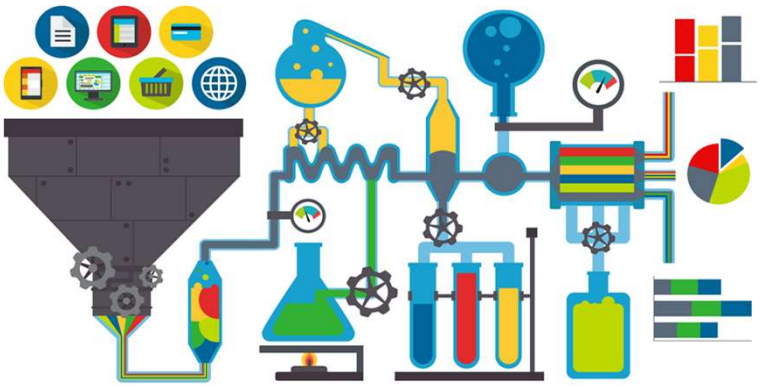

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Análisis exploratorio

Ciclo iterativo en el que se hacen preguntas acerca de los datos, y se responde usando técnicas de Visualización y Estadística**.**

1. **Define Project Objectives**
   - Specify business problem
   - Acquire subject matter expertise
   - Define unit of analysis and prediction target
   - Prioritize modeling criteria
   - Consider risks and success criteria
   - Decide whether to continue
2. **Acquire & Explore Data**
   - Find appropriate data
   - Merge data into single table
   - Conduct exploratory data analysis
   - Find and remove any target leakage
   - Feature engineering
3. **Model Data**
   - Variable selection
   - Build candidate models
   - Model validation and selection
4. **Interpret & Communicate**
   - Interpret model
   - Communicate model insights
5. **Implement, Document & Maintain**
   - Set up batch or API prediction system
   - Document modeling process for reproducibility
   - Create model monitoring and maintenance plan

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https://www.datarobot.com/wiki/machine-learning-life-cycle/

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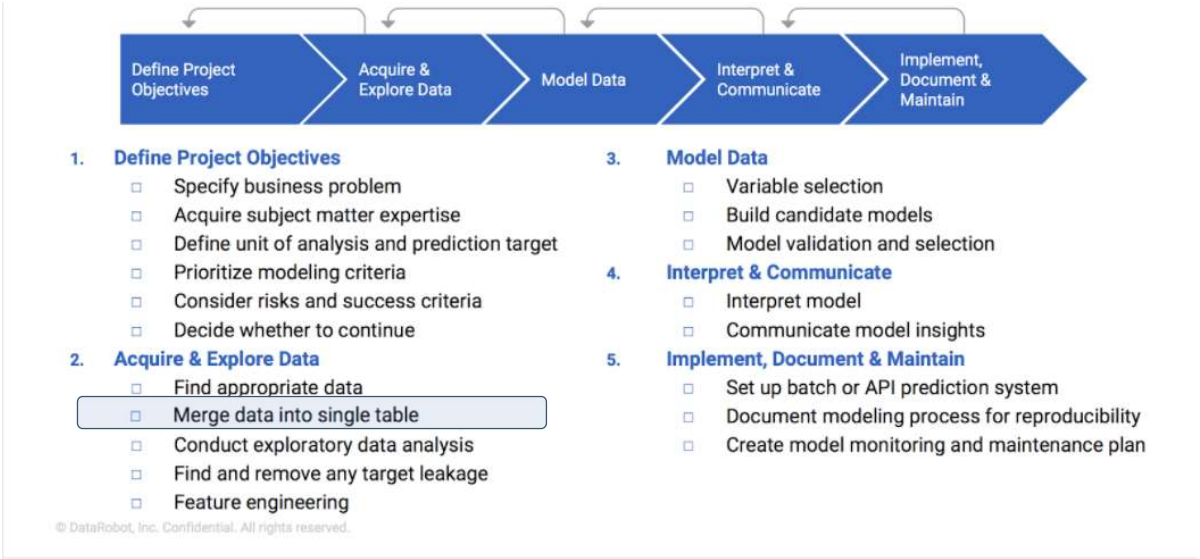

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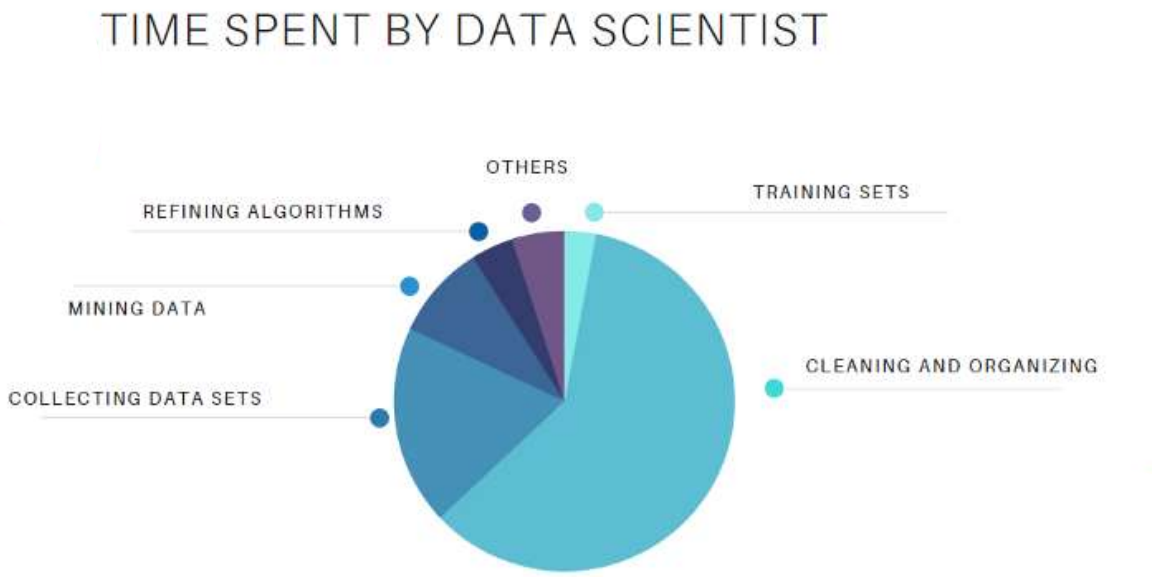
Importancia de los datos

- ¿La fuente de los datos es primaria o secundaria?
- ¿Puede haber sesgo en los datos?
- Almacenamiento y procesamiento (formato y cantidad de datos)



https://towardsdatascience.com/what-is-feature-engineering-importance-tools-and-techniques-for-machine-learning-2080b0269f10

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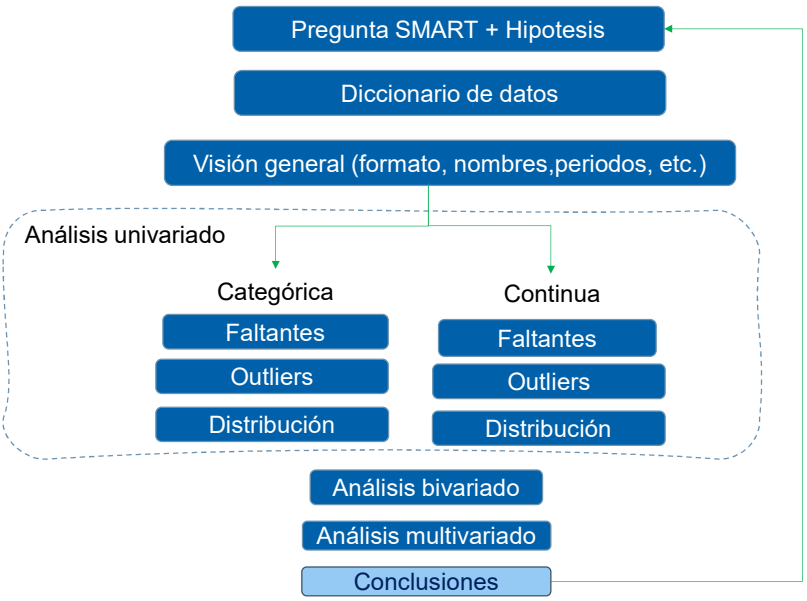
Flujo general



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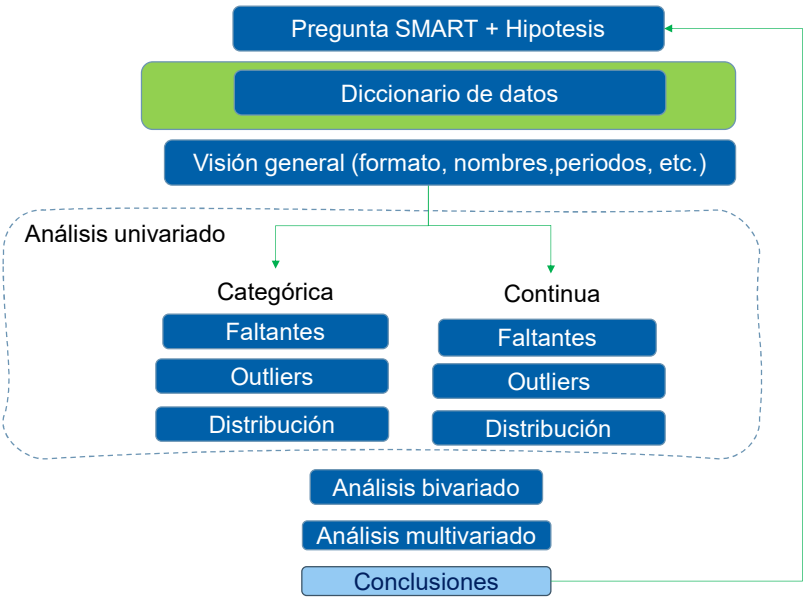


Diccionario de datos

Nombre de la variable	Descripción	Tipo de variable	Valores válidos
Age	Edad	Númerica	18 a 95
Child	Tiene hijos	Categórica	"Yes","No"
Num. Child	Número de hijos	Númerica	0 a 4
Job	Tipo de trabajo	Categórica	"admin.","unknown","unemployed","management","housemaid","entrepreneur","student", "blue-collar","self-employed","retired","technician","services"
marital	Estado Civil	Categórica	widowed
education	Nivel educativo alcanzado	Categórica	"unknown","secondary","primary","tertiary"
default	Tiene un crédito en mora	Categórica	"Yes","No"
balance	Balance promedio por año, en euros	Númerica	102127 a -8019
housing	Tiene préstamo para vivienda	Categórica	"Yes","No"
loan	Tiene préstamo personal	Categórica	"Yes","No"
contact	Tipo de contacto con el cliente	Categórica	"unknown","telephone","cellular"

Moro, R. et al.. Using Data Mining for Bank Direct Marketing. Proc. ESM'2011, pp. 117-121, 2011

Diccionario de datos

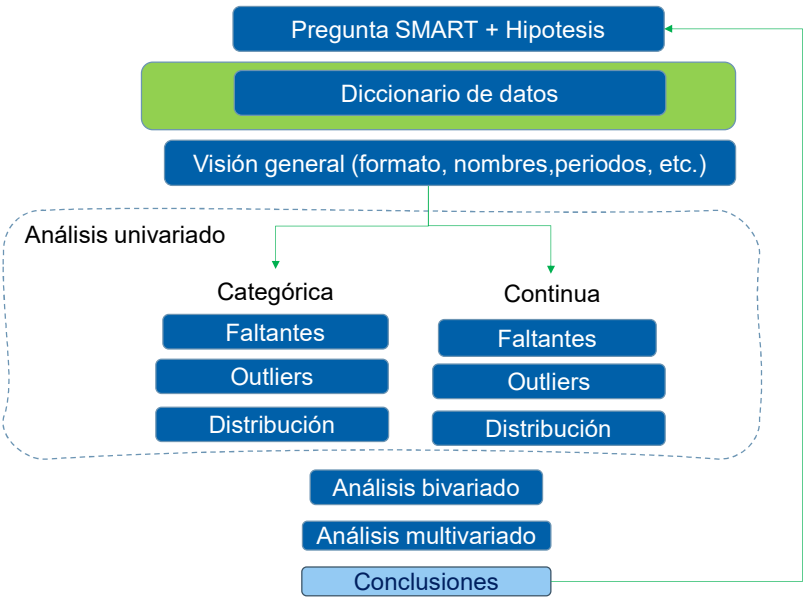
Variable: Nombre de la variable

Tipo: Tipo o formato de la variable. Esto puede ser categórico, numérico, booleano, etc.

Contexto: Información útil para entender el espacio semántico de la variable. En el caso de nuestro conjunto de datos, el contexto siempre es químico-físico, por lo que es fácil. En otro contexto, por ejemplo el inmobiliario, una variable podría pertenecer a un segmento en particular, como la anatomía del material o el social (¿cuántos vecinos hay?)

Expectativa: ¿Qué tan relevante es esta variable con respecto a nuestra tarea? Podemos usar una escala “Alto, Medio, Bajo”.

Flujo general



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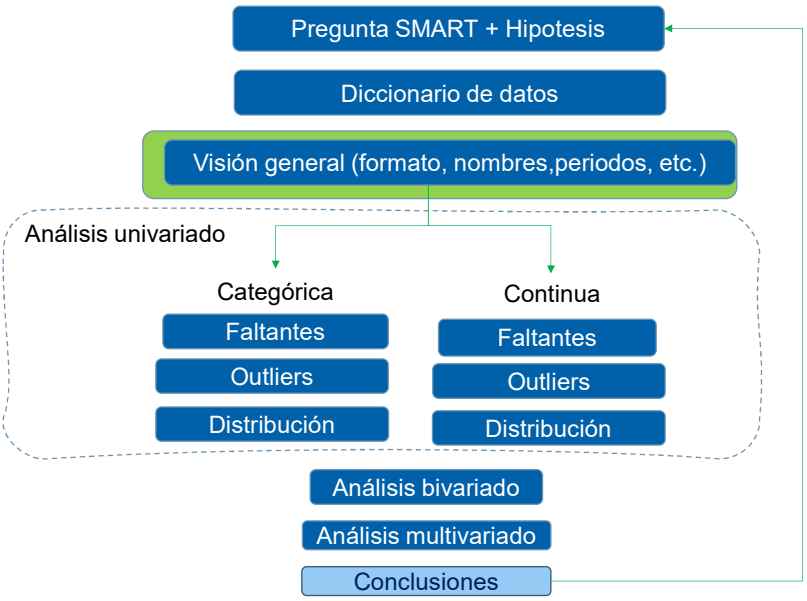
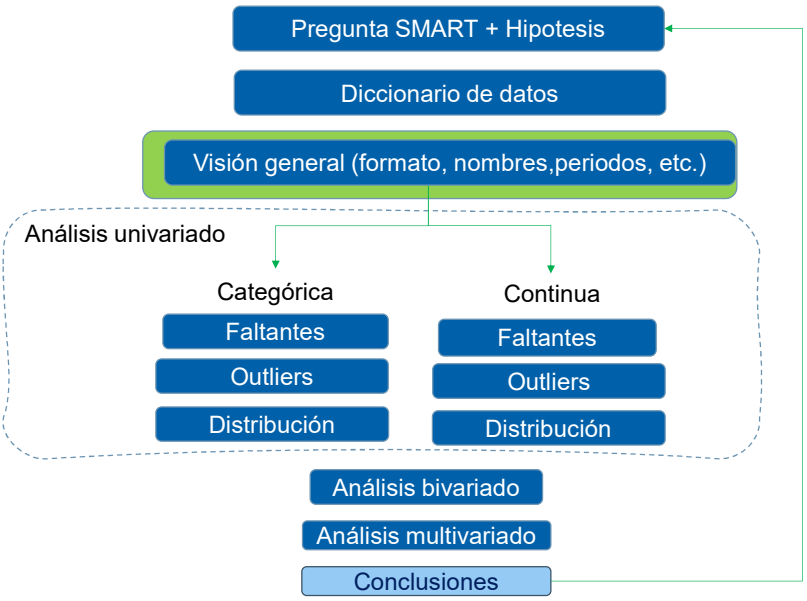


Vision general de los datos

1. Formato de los datos
2. Nombre de las columnas (Nomenclatura)
3. Tipos de datos
4. Unidades de medida
5. Variables redundantes o no útiles (IDs e información personal, data leakage)
6. Nuevas columnas necesarias

Flujo general



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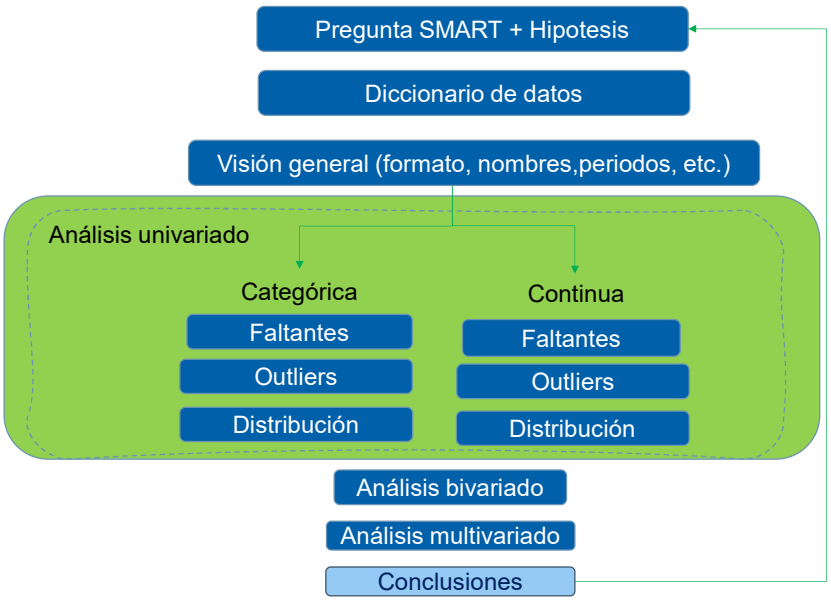


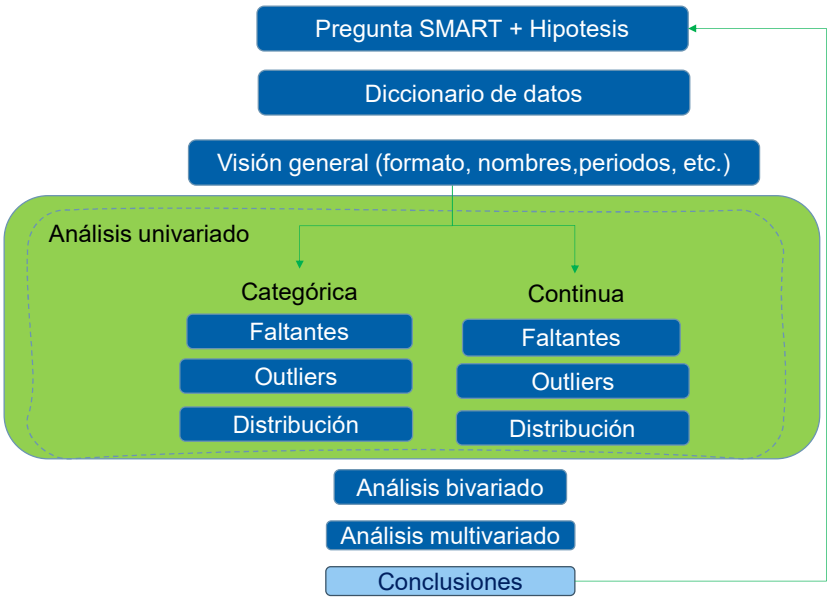
Problemas típicos

- Formato de las variables
- Observaciones duplicadas
- Valores perdidos
- Errores de digitación

Problemas típicos

- Valores inconsistentes, fuera de rango o inválidos
- Desconocimiento de los datos
- Data leakage

Flujo general



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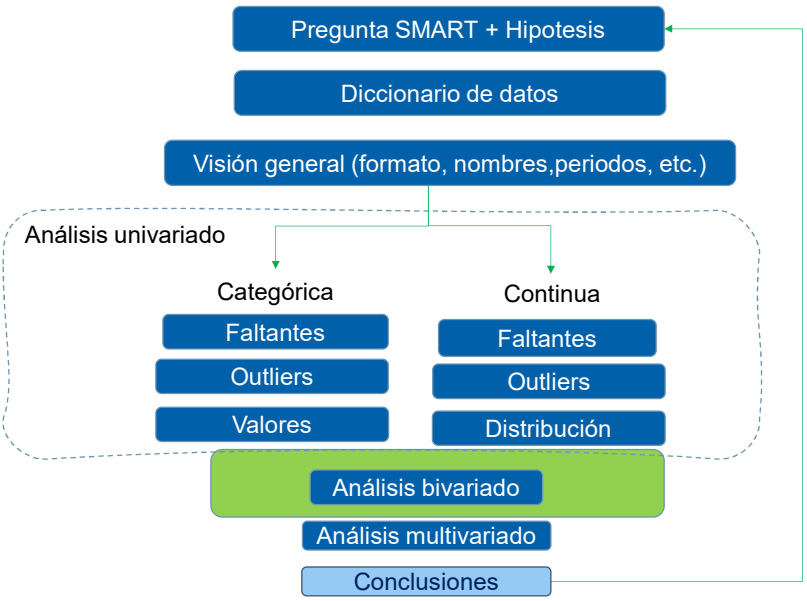

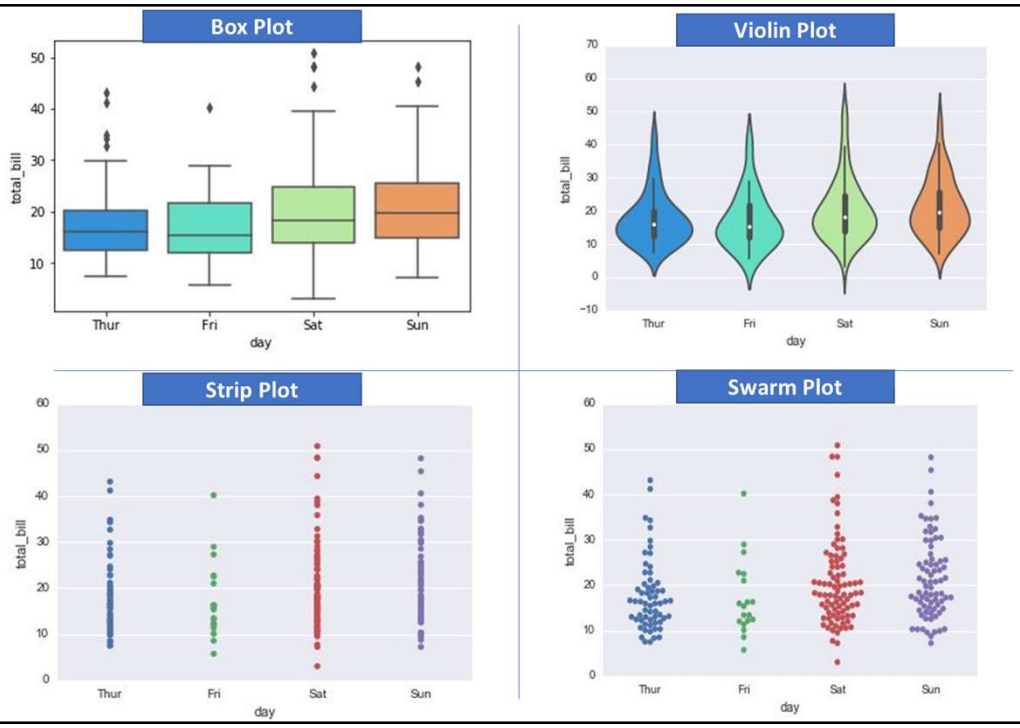

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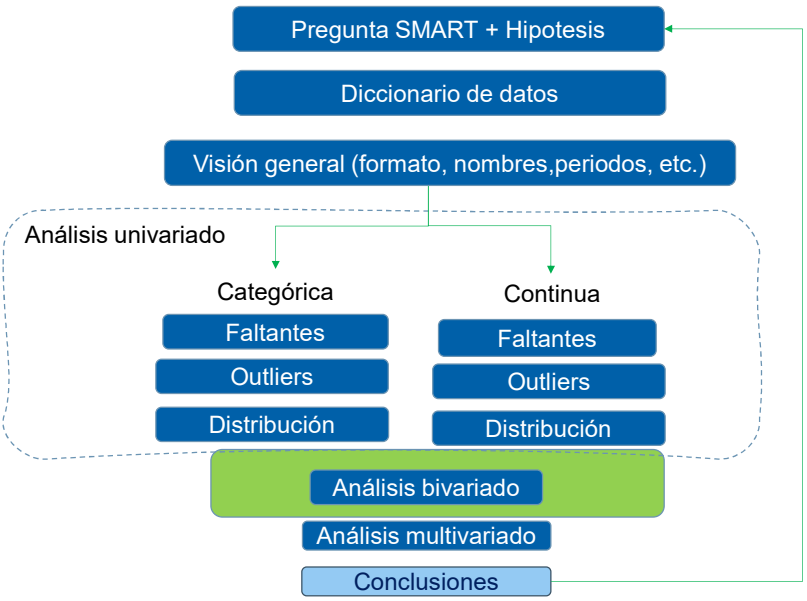

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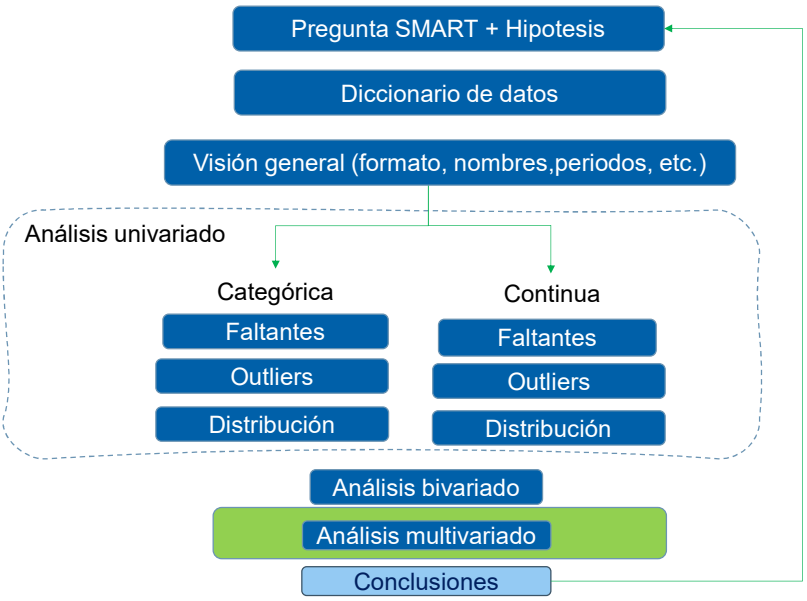
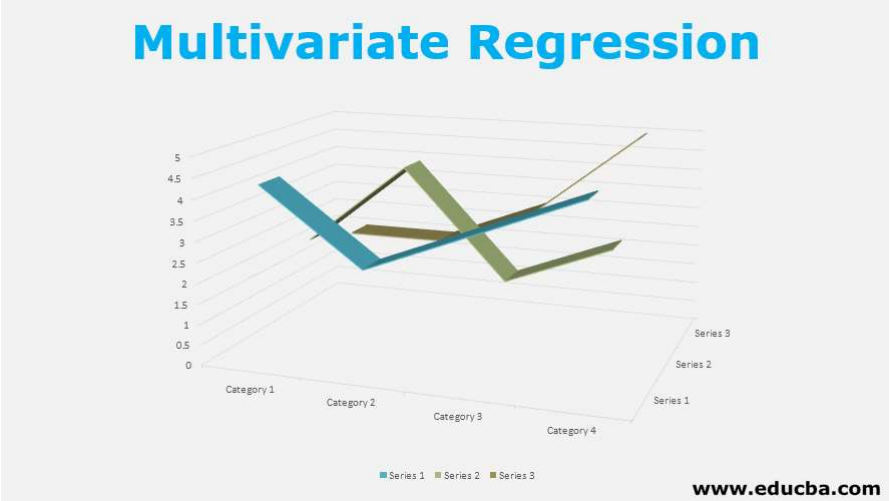
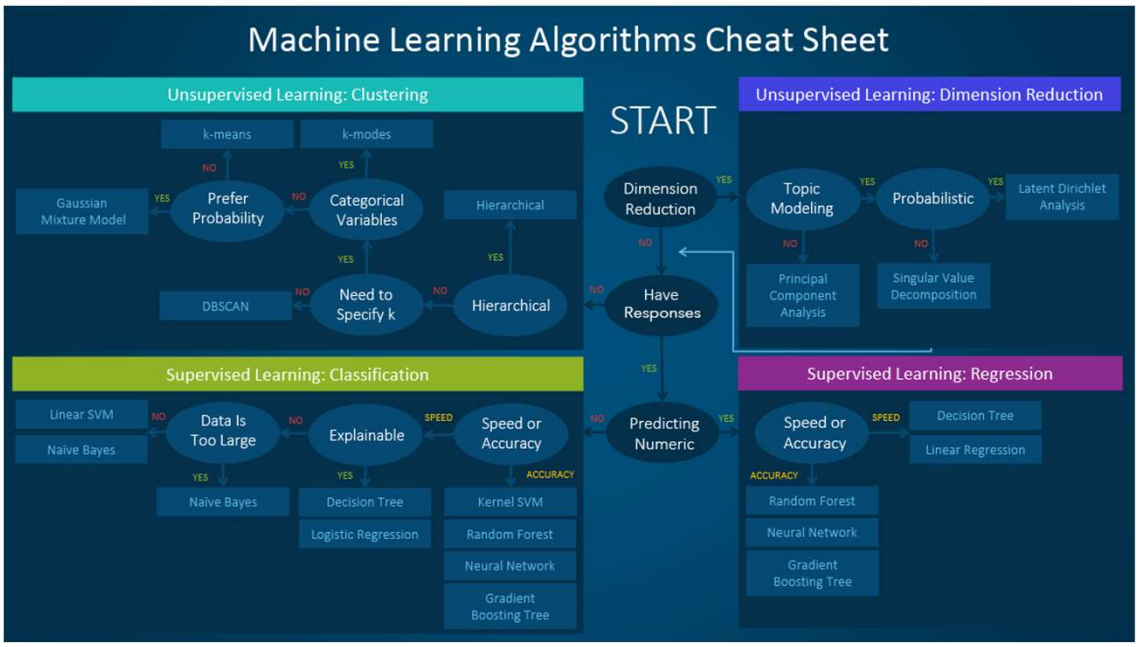




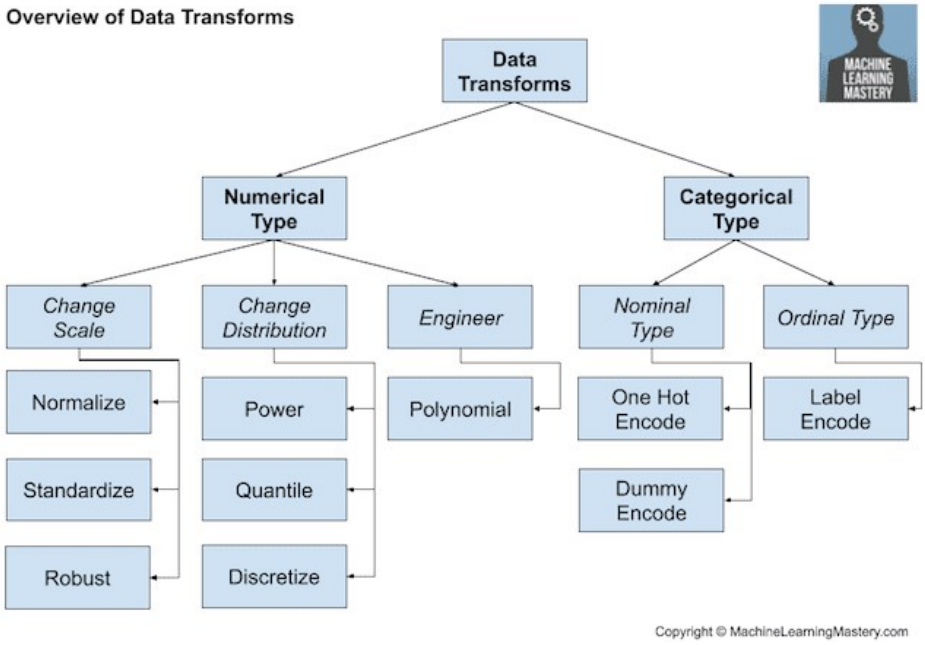
Preparación

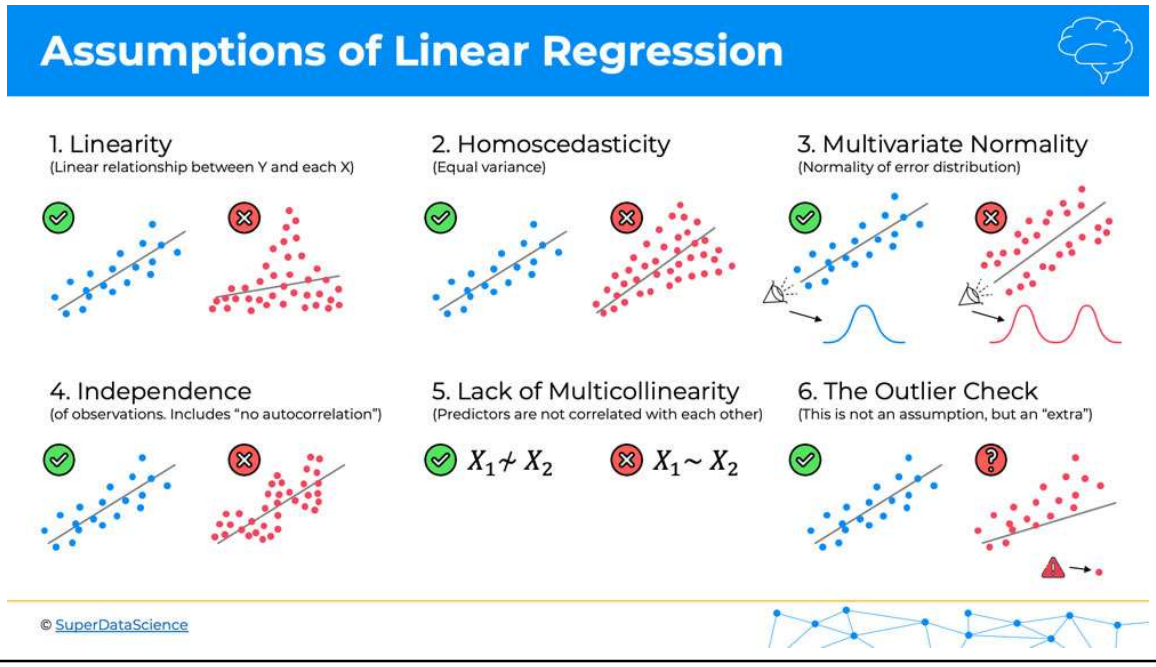
Técnicas para hacer que los datos puedan ser usados por los

Modelos de predicción (Numéricos, sin errores ni outliers, etc.)



https://blogs.sas.com/content/subconsciousmusings/2020/12/09/machine-learning-algorithm-use/





Análisis y preparación

Análisis exploratorio

Visión general (formato, nombres,periodos, etc.)

Categórica
- Faltantes
- Outliers
- Distribución

Continua
- Faltantes
- Outliers
- Distribución

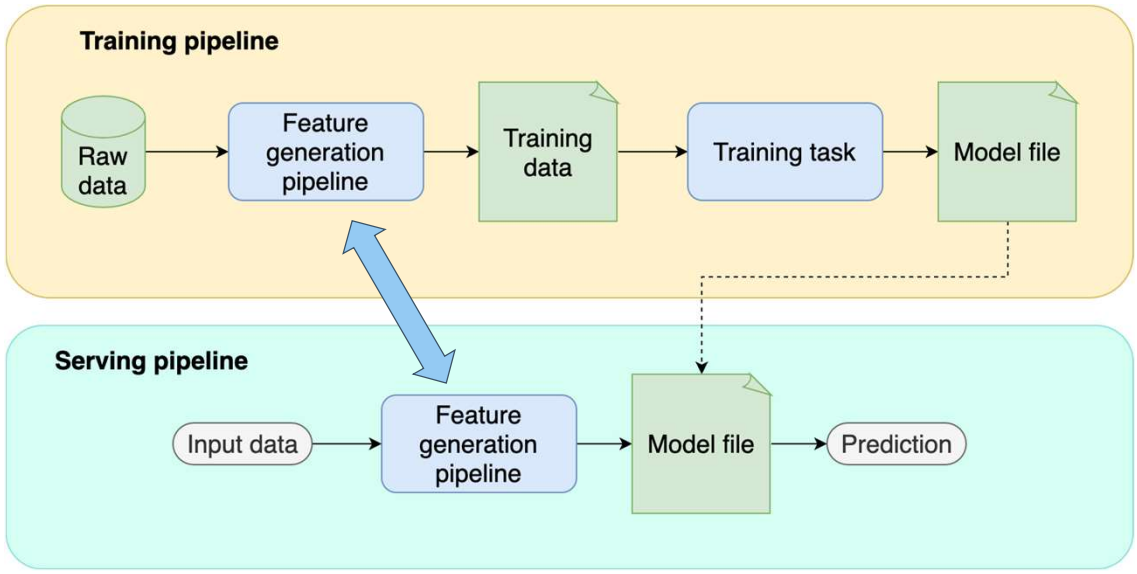
- Análisis bivariado
- Análisis multivariado
- Visualización

Preparación

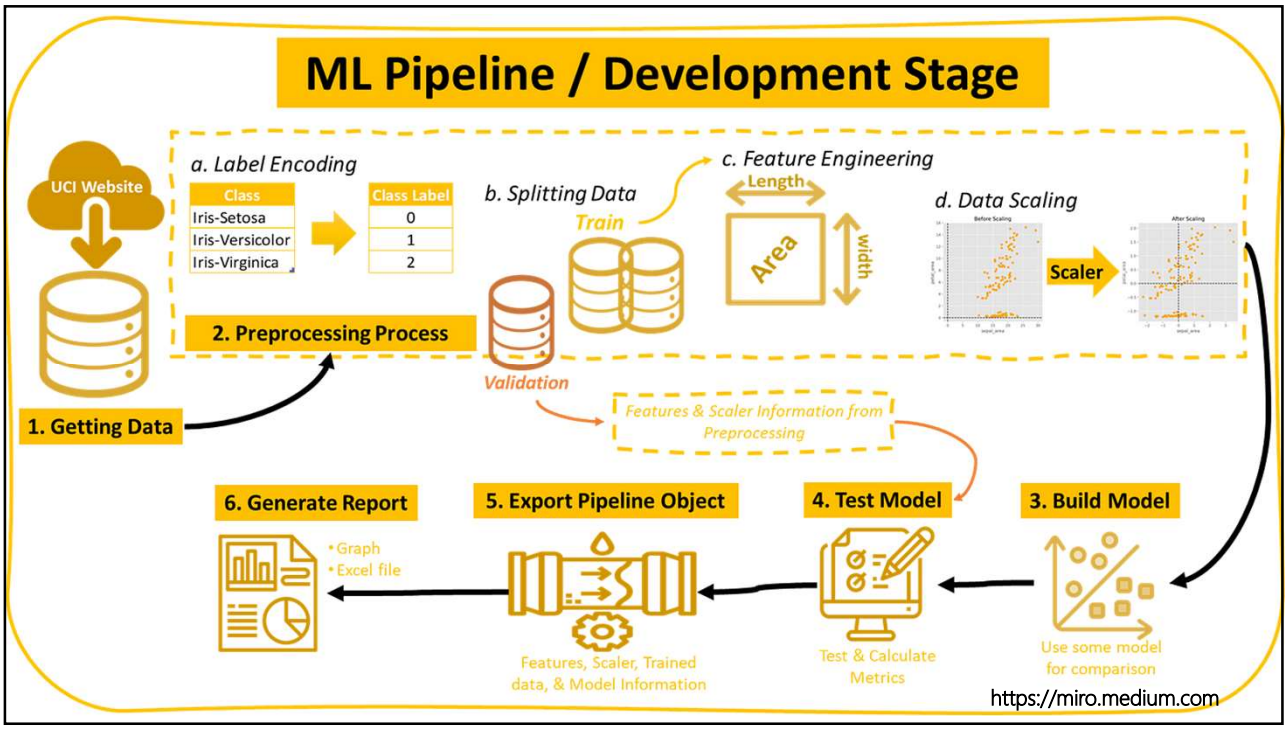
- Categórica <-> continua
- Normalización y estandarización
- Data leakage
- Feature engineering
- Feature selection

- Modelo

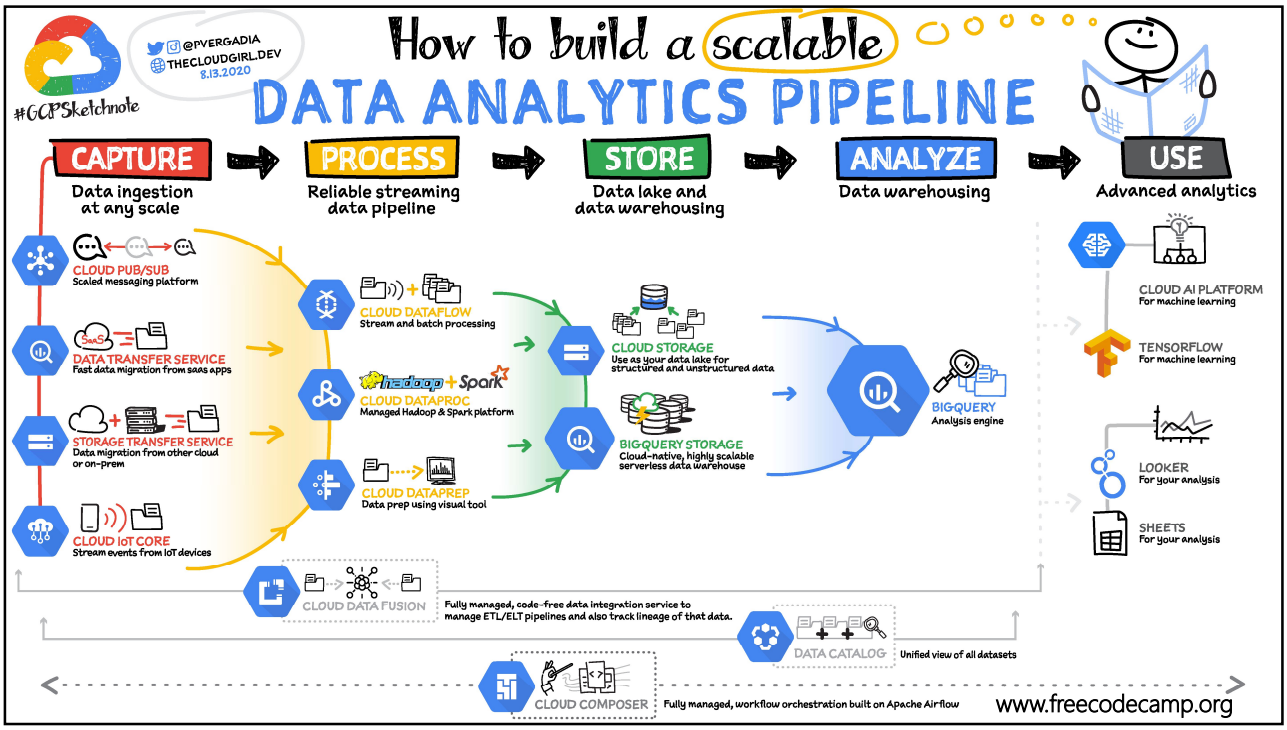
Feature generati
Pipeline



https://ploomber.io/blog/ml-testing-i/



https://miro.medium.com


#GCPSketchnote
@PVERGADIA
THECLOUDGIRL.DEV
8.13.2020
How to build a scalable
DATA ANALYTICS PIPELINE
CAPTURE
Data ingestion at any scale
PROCESS
Reliable streaming data pipeline
STORE
Data lake and data warehousing
ANALYZE
Data warehousing
USE
Advanced analytics
CLOUD PUB/SUB
Scaled messaging platform
DATA TRANSFER SERVICE
Fast data migration from saas apps
STORAGE TRANSFER SERVICE
Data migration from other cloud or on-prem
CLOUD IoT CORE
Stream events from IoT devices
CLOUD DATAFLOW
Stream and batch processing
CLOUD DATAPROC
Managed Hadoop & Spark platform
CLOUD DATAPREP
Data prep using visual tool
CLOUD STORAGE
Use as your data lake for structured and unstructured data
BIGQUERY STORAGE
Cloud-native, highly scalable serverless data warehouse
BIGQUERY
Analysis engine
CLOUD AI PLATFORM
For machine learning
TENSORFLOW
For machine learning
LOOKER
For your analysis
SHEETS
For your analysis
CLOUD DATA FUSION
Fully managed, code-free data integration service to manage ETL/ELT pipelines and also track lineage of that data.
DATA CATALOG
Unified view of all datasets
CLOUD COMPOSER
Fully managed, workflow orchestration built on Apache Airflow
www.freecodecamp.org

In
God
We
Trust